

Australian Probability Conference 2026

Program Schedule

22–25 June 2026

Talk format: Regular talks 30 min (25 min + 5 min Q&A). APSIG Lecture 60 min (50 min + 10 min Q&A).

Monday 22 June

Afternoon

Time	Speaker	Title / Event
12:30–13:00	<i>Registration and Welcome address</i>	
13:00–14:00	APSIG Lecture Kais Hamza (Monash)	<i>Are random random walks normal?</i>
Session 1 — Applied Probability and Mathematical Biology		
14:30–15:00	Han Gan (Waikato)	<i>Poisson-Dirichlet approximation of the inclusion process</i>
15:00–15:30	Jie Yen Fan (Monash)	<i>Estimation in age- and population-dependent models</i>
15:30–16:00	Ross McVinish (UQ)	<i>Lorenz ordering of parasite loads across hosts in Isham’s model</i>

Tuesday 23 June

Full day

Time	Speaker	Title
Session 2 — Lévy Processes and Finance		
9:00–9:30	Boris Buchmann (ANU)	<i>Weak subordination of multivariate Lévy processes</i>
9:30–10:00	Kevin Lu (ANU)	<i>Pricing energy spread options with variance gamma-driven Ornstein-Uhlenbeck dynamics</i>
10:00–10:30	Qingyuan Zhang (UQ)	<i>Continuous-time multi-armed bandits under random intervention times</i>
Session 3 — Stochastic Analysis and Applications		
11:00–11:30	Marco Carfagnini (Melbourne)	<i>Berry-Heisenberg random waves</i>
11:30–12:00	Shahid Khan (La Trobe)	<i>On anisotropic random fields on the sphere</i>
Session 4 — Random Structures and High-Dimensional Statistics		
13:30–14:00	Indrajit Jana (IIT Bhubaneswar)	<i>Spectrum of random centrosymmetric matrices</i>

Time	Speaker	Title
14:00–14:30	Radislav Vaisman (UQ)	<i>On the benefit of robust Bayesian confirmatory factor analysis</i>
14:30–15:00	Hongyi Lyu (Monash)	<i>Asymptotic power law of a random graph process</i>
Session 5 — Neural Networks and Uncertainty Quantification		
15:30–16:00	Joseph Wilson (UQ)	<i>Is the last layer sufficient for uncertainty quantification?</i>
16:00–16:30	Volter Entoma (UQ)	<i>A data-driven Fourier-mixture neural-network method for density estimation</i>
16:30–17:00	Yukan Perumal (UQ)	<i>Spouse-protected tontines: neural-network optimization for household decumulation</i>

Wednesday 24 June

Full day

Time	Speaker	Title
Session 6 — Inference for Stochastic Models		
9:00–9:30	Andriy Olenko (La Trobe)	<i>Estimation of Hurst functions of multifractional processes</i>
9:30–10:00	Adam Nie (ANU)	<i>Spurious factor analysis for high-dimensional functional time series</i>
10:00–10:30	David Warne (QUT)	<i>Multifidelity simulation and inference for non-Markovian reaction networks</i>
Session 7 — Applied Probability and Queueing		
11:00–11:30	Hritika Gupta (Wollongong)	<i>Transient waiting time distributions in call centres</i>
11:30–12:00	Angus Lewis (Adelaide)	<i>PH representations of exponentially tilted PH distributions</i>
Session 8 — Sequential Decision-Making and Reinforcement Learning		
13:30–14:00	Nan Ye (UQ)	<i>Scaling up sequential decision-making</i>
14:00–14:30	Joel Woodfield (UQ)	<i>The difficulty of learning simple policies in reinforcement learning</i>
14:30–15:00	Shahtaj Bano (RMIT)	<i>Multi-objective stochastic optimization of chemotherapy scheduling</i>
Session 9 — Multivariate Statistical Methods		
15:30–16:00	Rahul Singh (IIT Delhi)	<i>A tale of paired comparison models</i>
16:00–16:30	Biplab Paul (IIT Delhi)	<i>Clustering-based exact distribution-free k-sample tests for high-dimensional data</i>
16:30–17:00	Bahareh Ghanbari (RMIT)	<i>Tractable factor copula models</i>

Thursday 25 June

Morning

Time	Speaker	Title / Event
Session 10 — Machine Learning: Theory and Methods		
9:00–9:30	Jiading Liu (UQ)	<i>Learning kernel-based operators via random features</i>
9:30–10:00	Jonathan Wilton (UQ)	<i>Robust tree-based methods for learning with complementary labels</i>
10:00–10:30	Tran Nguyen Tung Doan (Griffith)	<i>Bridging incremental stochastic MM algorithms and mixture of experts models</i>
Session 11 — Extremes and Risk Analysis		
11:00–11:30	Chang Chen (UQ)	<i>Optimal decumulation strategy with quadratic variation as a risk measure</i>
11:30–12:00	Grace Burtenshaw (UQ)	<i>Exploring drivers of extreme housing price in Australia</i>
12:00–12:30	Meagan Carney (UQ)	<i>Runs of extremes in dynamical systems: scaling laws and applications</i>
12:30–13:00	<i>Closing remarks</i>	

Australian Probability Conference 2026

Abstracts

Monday 22 June

APSIG Lecture

Are random random walks normal?

Kais Hamza (Monash University)

Joint work with Laurent Tournier

Given a symmetric simple random walk $(X_n)_{n \geq 0}$, the family of symmetric simple random walks $(Y_n)_{n \geq 0}$ adapted to the filtration of $(X_n)_{n \geq 0}$ was studied in Collevocchio et al. (2022). In particular, the authors established necessary and sufficient conditions under which a suitably normalized two-dimensional process $((X_n, Y_n))_{n \geq 0}$ converges weakly to a two-dimensional Brownian motion. When this occurs, we say that the random walk is *normal*. In this paper, we investigate whether a “randomly selected” $(Y_n)_{n \geq 0}$ is normal. We consider a very general randomization procedure and look at both the quenched and annealed settings.

Session 1 — Applied Probability and Mathematical Biology

Poisson-Dirichlet approximation of the inclusion process

Han Gan (University of Waikato)

The finite inclusion process is an interacting particle system with N particles on L sites. In the so-called thermodynamic limit, where L is of the same order of N , the limiting stationary distribution for the process is the Poisson-Dirichlet distribution. Using a modification of [Gan & Ross, 2021], for any fixed N and L , we use Stein’s method to derive an explicit bound for the approximation error between the stationary distribution of the finite model and the Poisson-Dirichlet distribution. In the thermodynamic limit, this error is of order $1/N$.

Estimation in age- and population-dependent models

Jie Yen Fan (Monash University)

Joint work with Kais Hamza, Fima Klebaner, Ziwen Zhong

We consider general age- and population-dependent population systems, where individual birth and death rates depend not only on age but also on the overall population composition. These systems can be modelled as measure-valued stochastic processes. A Functional Law of Large Numbers and a Central Limit Theorem with large carrying capacity can be established. These results, with appropriate test functions, allow us to estimate the demographic rates and obtain some inference results.

Lorenz ordering of parasite loads across hosts in Isham’s model

Ross McVinish (The University of Queensland)

Nearly all parasite populations exhibit substantial variation in their distribution among hosts, a phenomenon known as aggregation. Isham proposed a model of parasite acquisition in which the host’s mortality rate increases linearly with parasite load. We show that, in Isham’s model, increasing parasite-induced host mortality increases parasite aggregation in the sense of the Lorenz order and related measures, when the distribution of the number of parasites entering a host at infectious contacts is log-concave. Furthermore, in the presence of parasite-induced host mortality, the parasite mortality rate need not have a monotone effect on aggregation.

Tuesday 23 June

Session 2 — Lévy Processes and Finance

Weak subordination of multivariate Lévy processes

Boris Buchmann (The Australian National University)

Joint work with Kevin Lu, Adam Nie

Subordination is the operation that evaluates a Lévy process at a subordinator, giving rise to a path-wise construction of a “time-changed” process. In probability semigroups, subordination was applied to create the variance gamma process, which is prominently used in financial modelling. However, subordination may not produce a Lévy process unless the subordinate has independent components or indistinguishable components. A new operation known as weak subordination is introduced that always produces a Lévy process by assigning the distribution of the subordinate conditional on the value of the subordinator, matching traditional subordination in law in the cases above. Weak subordination is applied to extend the class of variance-generalised gamma convolutions and to construct the weak variance-alpha-gamma process, which exhibits a wider range of dependence than traditional subordination.

Pricing energy spread options with variance gamma-driven Ornstein-Uhlenbeck dynamics

Kevin Lu (Australian National University)

Joint work with Tim Leung

We consider the pricing of energy spread options for spot prices following an exponential Ornstein-Uhlenbeck process driven by a sum of independent multivariate variance gamma processes, which gives rise to mean-reverting, infinite activity price dynamics. Within this class of driving processes, the Esscher transform is used to obtain an equivalent martingale measure with a focus on the weak variance alpha-gamma process. By deriving an analytical formula for the cumulant generating function of the innovation term, we obtain a pricing formula for forwards and apply an FFT method to price spread options. We also demonstrate how the model can be estimated

on energy prices under the real-world measure and calibrated on forward or call prices, and provide numerical results for Australian electricity market data.

Continuous-time multi-armed bandits under random intervention times

Qingyuan Zhang (The University of Queensland)

This paper examines multi-armed bandits in which actions are taken at random discrete times. The model consists of independent arms. When an arm is operated, it must remain active for a random duration, modelled by the inter-arrival time of a (possibly arm-dependent) renewal process. For arms evolving as a Lévy process, we provide an explicit characterisation of the Gittins index, which is known to yield an optimal strategy. Furthermore, when the inter-arrival times are exponential and the arms evolve as either a spectrally negative Lévy process, a reflected spectrally negative Lévy process, or a diffusion process, the Gittins index is explicitly characterised in terms of the scale function or diffusion characteristics, respectively. Numerical experiments are performed to support the theoretical results.

Session 3 — Stochastic Analysis and Applications

Berry-Heisenberg random waves

Marco Carfagnini (The University of Melbourne)

Joint work with Anna Paola Todino

In the 1970s, Berry argued that in the high-energy limit wave functions locally look like random superpositions of independent plane waves, all having the same wavenumber. He introduced a Gaussian random field whose sample paths are generalised Laplace eigenfunctions. The aim of this talk is to present a similar model for the sub-Laplacian on the Heisenberg group, which is the analogue of the Euclidean space in sub-Riemannian geometry. This is based on a current project with Anna Paola Todino from the University of Eastern Piedmont, Italy.

On anisotropic random fields on the sphere

Shahid Khan (La Trobe University)

Joint work with Andriy Olenko, Claudio Durastanti

This talk addresses fundamental questions about the existence and structure of isotropic and anisotropic spherical random fields and the properties of composite transformations of isotropic fields. Using geometric approaches, it introduces a hierarchical structure comprising several classes of anisotropic random fields. We provide a geometric and spectral classification of second-order anisotropy on the sphere and establish connections between geometric invariance properties and the structure of the angular power spectra of spherical random fields. Three composite models (multiplicative, convolutional, and deformed random fields) built on simpler isotropic components are discussed. The results lay the foundations for applications in statistics, guiding feasible models, their spectral representations, and appropriate sampling schemes for anisotropic spherical data.

Spectrum of random centrosymmetric matrices

Indrajit Jana (Indian Institute of Technology, Bhubaneswar)

Joint work with Sunita Rani

We analyse the asymptotic fluctuations of linear eigenvalue statistics of random centrosymmetric matrices with i.i.d. entries. We prove that for a complex analytic test function, the centred and normalised linear eigenvalue statistics converge to a normal distribution. We find the exact expression of the variance of the limiting normal distribution via combinatorial arguments. Moreover, we argue that the limiting spectral distribution of properly scaled centrosymmetric matrices follows the circular law.

On the benefit of robust Bayesian confirmatory factor analysis

Radislav Vaisman (The University of Queensland)

Joint work with Mitchell Scovell, Nikolai Kinaev, Javier Fernandez

Data normality is a typical assumption when performing confirmatory factor analysis, but this assumption frequently fails in practice. We construct an efficient sampler for a robust Bayesian confirmatory factor analysis using the Student's t distribution and provide a novel approach for calculating the corresponding marginal likelihood. The proposed strategy is both computationally efficient and performs favourably compared to standard approaches that rely on the normality assumption.

Asymptotic power law of a random graph process

Hongyi Lyu (Monash University)

Joint work with Andrea Collevecchio, Nick Wormald

We consider a random graph evolution model proposed by Richards and Wormald (2014) defined as follows: at each time step a new vertex is added to the graph by first joining to an existing vertex u and then a neighbour of u , where both choices are made uniformly at random. It has been conjectured that the degree distribution obeys a power law asymptotically. We study the neighbourhood growth of each vertex v by coupling it with a randomly evolving tree. As a result, we obtain power-law upper and lower bounds on the asymptotic degree distribution of the original graph model.

Is the last layer sufficient for uncertainty quantification?

Joseph Wilson (The University of Queensland)

Joint work with Fred Roosta, Liam Hodgkinson, Chris van der Heide

Epistemic uncertainty quantification (UQ) for deep neural networks (DNNs) is important for the safe adoption of AI in mission-critical settings. Several leading UQ methods linearize trained DNNs to form Bayesian generalized linear models, where a common approximation is to linearize

only the final layer rather than the full network. We compare these models through the lens of random matrix theory, analysing the predictive covariance structures induced by full-network and last-layer linearization. Our analysis suggests that full-network linearization does not provide a clear theoretical advantage for epistemic UQ over its last-layer counterpart. These findings are supported by large-scale empirical evaluation across modern machine learning tasks, where last-layer approximations achieve comparable UQ performance with substantially improved computational efficiency.

A data-driven Fourier-mixture neural-network method for density estimation

Volter Entoma (The University of Queensland)

Joint work with Duy-Minh Dang

We propose a data-driven Fourier-trained neural-network method for estimating fixed-horizon probability densities from empirical characteristic function (CF) information. The estimator is a positive Gaussian-Laplace mixture with a closed-form CF, so training is performed directly in Fourier space while preserving nonnegativity and unit mass. For the direct i.i.d. sampling setting we derive an expected L^2 error bound separating Fourier truncation, training, discretization, and CF sampling errors; for the resampling-based pseudo-sampling setting we obtain a conditional analogue with two additional pseudo-law discrepancy terms. Numerical experiments show competitive performance on Gaussian-mixture benchmarks, clear gains on heavy-tailed targets, L^2 error decay consistent with theory, and effective estimation of one-year Australian equity return distributions from resampled dependent data.

Spouse-protected tontines: neural-network optimization for household decumulation

Yukan Perumal (The University of Queensland)

Joint work with Duy-Minh Dang

Motivated by retirement products that continue income to a surviving spouse, we study household decumulation in a spouse-protected tontine. The household is represented by a four-state survival process for retiree and spouse, and the tontine gain rate is state-dependent because the contract exits the pool only when both members have died. The resulting finite-horizon control problem trades off expected cumulative inflation-adjusted household payments against the CVaR of a state-contingent terminal shortfall in a multi-asset portfolio setting. Feedforward neural networks approximate the withdrawal and rebalancing controls. To quantify the actuarial cost of survivor payments, we introduce a policy-induced spouse-continuation load computed ex post from the optimised spouse-only payment stream. Numerical experiments with long-horizon Australian data are conducted.

Wednesday 24 June

Session 6 — Inference for Stochastic Models

Estimation of Hurst functions of multifractional processes

Andriy Olenko (La Trobe University)

Joint work with A. Ayache (University of Lille), N. Samarakoon (La Trobe University)

This talk presents results in statistical inference for multifractional stochastic processes, focusing on two models: the Riemann–Liouville fractional Brownian motion and the recently introduced Gaussian Haar-based multifractional process. These models extend classical fractional Brownian motion by replacing the constant Hurst parameter with deterministic Hurst functions, allowing for modelling of signals with time-varying local regularity. A key challenge is the estimation of local Hölder regularity from discretely observed data under infill asymptotics, where the observation grid becomes increasingly dense over a fixed time interval. We study pointwise estimators of the Hurst function for both models via a fine analysis of generalised quadratic variations and second-order increments of the processes, and establish consistency. Numerical examples are provided.

Spurious factor analysis for high-dimensional functional time series

Adam Nie (The Australian National University)

Joint work with Yanrong Yang, Hanlin Shang, Yi He

This article explores a general factor structure for high-dimensional nonstationary functional time series, encompassing a wide range of factor models studied in the existing literature. We investigate the asymptotic spectral behaviour of the sample covariance operator and establish a novel sufficient condition, formulated in terms of a new effective rank, under which empirical eigenanalysis yields spurious results. This generalises the results of Onatski and Wang (*Econometrica*, 2021) from high-dimensional time series to the functional framework. Contrary to the high-dimensional time series setting, models with only a small number of strong non-stationary factors may still produce spurious limits in the functional framework. Simulation studies and real data analysis on age-specific mortality rates from multiple locations illustrate the findings.

Multifidelity simulation and inference for non-Markovian reaction networks

David Warne (Queensland University of Technology)

Joint work with Thomas Steele

Stochastic models of biochemical reaction networks are widely used to capture intrinsic noise in cellular systems. However, biological processes such as gene transcription and translation introduce history-dependent dynamics requiring non-Markovian processes. We develop efficient non-Markovian simulation algorithms for well-mixed stochastic biochemical reaction networks with state- and time-dependent delays, generalising the next reaction method and τ -leaping to support arbitrary inter-event time distributions. We also introduce a coupling scheme to generate exact non-Markovian sample paths correlated to approximate τ -leaping paths, enabling substantial computational gains for Bayesian inference through multifidelity simulation-based inference. The approach is demonstrated on a gene regulation model with delayed autoinhibition, achieving improvements of two orders of magnitude in both simulation accuracy and inference efficiency.

Session 7 — Applied Probability and Queueing

Transient waiting time distributions in call centres

Hritika Gupta (University of Wollongong)

Joint work with Peter Taylor, Mark Fackrell

We present a method to calculate the time-dependent waiting time distributions for call centres that employ skills-based routing. Many call centres operate under service level agreements requiring targets such as answering 90% of calls within 20 seconds. We model the call centre as a continuous-time Markov chain and use Laplace transform techniques to compute the relevant performance measures. We also demonstrate the use of this method to find the optimal routing policy in a certain class of policies.

PH representations of exponentially tilted PH distributions

Angus Lewis (University of Adelaide)

A phase-type (PH) distribution is the distribution of time until absorption of a continuous-time Markov chain (CTMC) with one absorbing state. If a distribution has density f , the exponentially tilted density is proportional to f multiplied by an exponential function. Via the Laplace transform one can show that if f is the density of a PH distribution then the tilted density is also PH. However, this does not yield a PH representation, i.e., a description in terms of time until absorption of a CTMC. This talk discusses how to construct such representations and thereby recover a CTMC interpretation. The case where the tilting exponential has a positive rate is particularly rich, utilising infinite-dimensional PH distributions and time reversal. Applications include exact simulation of tilted densities in extreme value theory.

Session 8 — Sequential Decision-Making and Reinforcement Learning

Scaling up sequential decision-making

Nan Ye (The University of Queensland)

Sequential decision-making is ubiquitous, but often computationally intractable due to challenges including uncertain dynamics, partial observability, and the interdependence of decisions at different time steps. In this talk, I will discuss some recent advances in scaling up solution algorithms for a general mathematical framework called Partially Observable Markov Decision Processes.

The difficulty of learning simple policies in reinforcement learning

Joel Woodfield (The University of Queensland)

Reinforcement learning (RL) algorithms can find well-performing policies for many problems, but the resulting policies are often complex and difficult to interpret. In contrast, simple policies often exist, but can be surprisingly difficult to learn using standard RL methods. In this talk, we explore why this occurs and the challenges of learning interpretable policies in RL.

Multi-objective stochastic optimization of chemotherapy scheduling

Shahtaj Bano (RMIT University)

Optimizing chemotherapy treatment planning requires balancing multiple conflicting objectives under significant uncertainty. We propose a stochastic model that jointly optimizes treatment cost, side effects, tumour reduction, drug intensity, number of active drugs, and relapse risk, incorporating nonlinear tumour dynamics and patient-specific variability across scenarios. We consider the two-stage case where the practitioner administers a treatment and then decides on recourse actions based on the patient's response. A decomposition-based solution approach combining Progressive Hedging with a Frank–Wolfe inner approximation is introduced to handle nonlinear dynamics and scenario uncertainty. The algorithm produces a sequence of solutions with a nondecreasing dual lower bound, enabling solution quality assessment via the primal–dual gap.

Session 9 — Multivariate Statistical Methods

A tale of paired comparison models

Rahul Singh (Indian Institute of Technology, Delhi)

Joint work with George Iliopoulos, Ori Davidov

Paired comparison models are central to ranking and preference analysis. We examine how graph-theoretic structures naturally emerge in representing paired comparison data. We then introduce the cardinal paired comparison model and present its main theoretical properties, both without and with covariates, using graph-based conditions in the fixed-item setting. Extensions of the model to infinitely many items are also discussed, along with numerical experiments that highlight the behaviour of the estimators. Finally, we compare the proposed model with the celebrated Bradley–Terry model.

Clustering-based exact distribution-free k -sample tests for high-dimensional data

Biplab Paul (Indian Institute of Technology, Delhi)

Joint work with Shyamal K. De, Anil K. Ghosh

Testing homogeneity of k ($k \geq 2$) multivariate distributions is challenging when the dimension greatly exceeds the sample size (HDLSS regime). We propose nonparametric tests with the distribution-free property in finite samples, based on a high-dimensional clustering algorithm that partitions the data into a contingency table. Test statistics are constructed from the cell frequencies of this table. Under appropriate regularity conditions, we prove consistency in the HDLSS asymptotic regime. A multiscale approach aggregating results over different numbers of partitions is also considered. Extensive simulations and benchmark datasets illustrate superiority over existing methods.

Tractable factor copula models

Bahareh Ghanbari (RMIT University)

Joint work with Pavel Krupskiy, Laleh Tafakori, Yan Wang

Gaussian factor models are widely used to characterise multivariate dependence but are limited in high-dimensional settings by restrictive tail behaviour. We propose a novel class of factor copula models combining the tractability of Gaussian factor models with the flexibility of copula-based dependence modelling. Each observed variable is represented as a mixture of a latent Gaussian

factor and a residual component with random loadings; residual dependence is introduced through an exchangeable one-factor copula, allowing flexible tail and asymmetric dependence. Both continuous and discrete specifications of the random loadings lead to analytically tractable correlation matrices, enabling fast least-squares parameter estimation. We establish that the latent factor can be consistently recovered as both sample size and dimension increase, and demonstrate the model's performance in high-dimensional simulation studies.

Thursday 25 June

Session 10 — Machine Learning: Theory and Methods

Learning kernel-based operators via random features

Jiading Liu (The University of Queensland)

Joint work with Lei Shi, Hien Duy Nguyen, Xin Guo

Learning operators between function spaces from data has become a fundamental paradigm in many applications and theoretical studies. A typical example is learning the Green's function of a linear partial differential equation (PDE). We introduce a novel data-driven kernel framework for learning operators that incorporates the random feature approach and handles noisy, discretely observed functional data. Rigorous theoretical guarantees for performance are provided.

Robust tree-based methods for learning with complementary labels

Jonathan Wilton (The University of Queensland)

Joint work with Nan Ye

Complementary labels are a type of weak supervision in classification problems where only information about classes that each example does *not* belong to is available. Despite the success of tree-based models in fully supervised classification, existing algorithms for the complementary labels problem generally use neural networks. This talk introduces a decision tree algorithm for complementarily labelled data whose predicted class distributions at each node are unbiased estimators of the true label distributions. Random forest ensembles of these trees achieve strong performance on tabular data, consistently outperforming neural network methods without hyperparameter tuning, and provide robust feature importance scores comparable to those of fully supervised trees.

Bridging incremental stochastic MM algorithms and mixture of experts models

Tran Nguyen Tung Doan (Griffith University)

Joint work with Trung Khang Tran, Trung Tin Nguyen, Gersende Fort, Hien Duy Nguyen, Binh T. Nguyen, Florence Forbes, Christopher Drovandi

Streaming data renders traditional batch algorithms in statistics and machine learning impractical, driving the need for incremental stochastic estimation methods. Softmax-gated Mixture of Experts

(MoE) models are particularly relevant in this context. While incremental stochastic EM and Majorization-Minimization (MM) algorithms exist, they are not applicable to softmax-gated MoE models. We develop a novel incremental stochastic MM algorithm specifically for these models, yielding explicit incremental updates with consistency guarantees under verifiable assumptions. In numerical tests, the proposed algorithm outperforms standard stochastic optimizers including SGD, RMSProp, Adam, and Sophia across both synthetic and real datasets.

Session 11 — Extremes and Risk Analysis

Optimal decumulation strategy with quadratic variation as a risk measure

Chang Chen (The University of Queensland)

Joint work with Duy-Minh Dang

We determine the optimal decumulation strategy with quadratic variation as the risk measure. Our objective is to maximise expected cumulative withdrawals while minimising wealth fluctuation, granting investors better control over risk throughout the investment horizon. We adopt the Annually Recalculated Virtual Annuity (ARVA) spending rule to avoid irrational withdrawal behaviour. A data-driven neural network approach determines the optimal asset allocation under realistic constraints such as no leverage and discrete rebalancing. Compared to a constant weight strategy with the same expected cumulative withdrawals, our strategy substantially reduces wealth fluctuation under both parametric market models calibrated from historical data and bootstrapped simulations, as well as multi-asset international diversification scenarios.

Exploring drivers of extreme housing price in Australia

Grace Burtenshaw (The University of Queensland)

Joint work with Meagan Carney, Ashley Burtenshaw

In recent years Australia has observed a growing, unexplained resilience of increasing house price trends. We construct a differential equation model of house price to develop intuition for its historical behaviour and responsiveness to changes in mortgage rates. Using this model, we identify a point of decoupling between house price and mortgage rate, with supply limitations identified as the main driver. Modern extreme value techniques are then applied to real-world data to investigate how the effectiveness of mortgage rate in moderating extreme house prices has changed before and after this decoupling. We find that without an increase in the housing supply chain, an 11% increase in mortgage rate will be needed to slow extreme housing costs.

Runs of extremes in dynamical systems: scaling laws and applications

Meagan Carney (The University of Queensland)

Joint work with Matthew Nicol, Mark Holland, Phuong Tran, Ruethaichanok Kardkasem, Grace Burtenshaw

Catastrophic events such as floods, heat waves, and surges in energy demand rarely arise from a single extreme observation but from sustained runs of extremes. Classical extreme value techniques model isolated maxima well but lose power in this regime, as requiring several consecutive extremes sharply depletes available tail data. We derive scaling laws relating the limiting distribution of run-based maxima to that of the original observable under standard

weak dependence assumptions, spanning the Fréchet, Weibull, and Gumbel domains. These results reveal an extension of max-stability: the shape parameter is preserved across all run lengths, while the location and scale follow explicit dynamical scaling laws. The approach is illustrated through case studies on catastrophic precipitation events across eastern Australia and gas demand extremes in Australian energy markets.
